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Leopoldo Soto

The main issues of research on dense magnetized plasmas in PF discharges

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Abstract

This invited lecture presents the main physical problems met during studies of dense ($> 10^{16} \text{ cm}^{-3}$) plasma influenced by strong magnetic fields, i.e. when ion cyclotron motions are weakly disturbed by electron–ion collisions. Examples of high-current pulsed discharges of the Z-pinch and plasma-focus (PF) type, which can produce dense magnetized plasma (DMP) are presented. The paper concerns mainly PF-type discharges. Particular attention is paid to different phases of such discharges. The breakdown and formation of a dense plasma layer (current sheath), its axial acceleration and the final radial collapse phase are analyzed. The most important physical phenomena are identified, and some theoretical models are described. The main issues of experimental and theoretical studies of DMP in PF-type facilities are shown.

(Some figures in this article are in colour only in the electronic version)

1. Introduction

Low-temperature plasma (at the electron temperature $T_e < 5 \text{ eV}$) constitutes a mixture of free electrons, ions and neutral atoms, while high-temperature (with $T_e > 10 \text{ eV}$) plasma may be completely ionized, forming a mixture of electrons and nuclei. Electrical and magnetic fields can interact with plasma, and when an influence of a magnetic field is so strong that the ion cyclotron frequency ($\omega_{ci} = Z_i e H / m_i c$) is higher than the frequency of electron–ion collisions ($\nu_{ei} = 25 n_i Z_i^2 / T_e^{3/2}$), such plasma becomes magnetized [1].

In so-called dense plasmas the electron concentration (n_e) is higher than 10^{16} cm^{-3} , and it is relatively difficult to magnetize such plasmas, in which collisions occur very often. Nevertheless, one can observe dense magnetized plasmas (DMP) in nature, e.g. within solar flares as well as in lightnings in the earth atmosphere, as shown in figure 1.

Studies of DMP phenomena in nature have been limited to theoretical modeling because the experimental conditions are irreproducible and they cannot be controlled. Therefore, numerous efforts have been undertaken to produce DMP in different laboratories. In fact, DMP can also be produced in the laboratory by high-current pulsed discharges, e.g. those of the Z-pinch or plasma-focus (PF) type.

In fact the oldest method used in order to generate high-temperature DMP has been based on high-current



Figure 1. Solar flares emitted from the Sun surface, which shows relatively dense and hot plasma streams driven by very strong magnetic fields (on the left), and the intense lighting, in which some portions of the current channels contain dense (10^{16} – 10^{19} cm^{-3}) plasma influenced by strong magnetic fields of high currents of the atmospheric discharge.

pulsed discharges between metal electrodes. The simplest configuration was a set of two conical, hemispherical or plane electrodes placed at a chosen distance inside a vacuum chamber. After filling the chamber with a working gas or gas mixture (usually under a relatively low pressure) the discharge was initiated by the application of a high-voltage pulse between two electrodes. If the electrodes are adjusted along the common z -axis the discharge can form a plasma

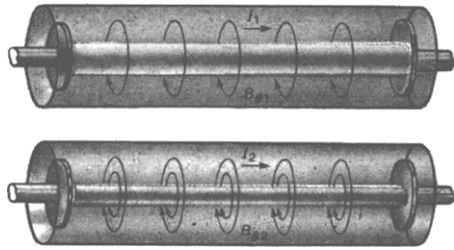


Figure 2. Schematic of the Z-pinch experiment, which shows the formation and compression of a plasma column by the azimuthal magnetic field generated by the discharge current.

column, which could undergo self-compression (Z-pinch) by its own magnetic field, as shown in figure 2.

Another important electrode configuration was a system consisting of two coaxial electrodes of different diameters, separated by a tubular glass or a ceramic insulator embracing the inner electrode at its beginning. Such electrodes were used in so-called coaxial plasma guns [2] as well as in PF devices of the Filippov- or Mather-type [3–6].

In the coaxial plasma guns of the Marshall type a working gas was injected into the inter-electrode region by a fast-acting valve, placed usually in the middle of the inner electrode. After forming a ring-shaped gas-cloud a high-voltage pulse was applied between the electrodes to initiate the plasma discharge. Such coaxial plasma guns could produce pulsed plasma streams (called often as plasma jets) and they were used as plasma injectors in many laboratory and space experiments [2, 7].

In contrast to the conventional coaxial guns, typical PF-type devices are usually initiated under a static initial pressure considerably higher than that in the coaxial injectors. A high-current pulse discharge, performed under such conditions, can form a dense plasma layer (a current sheath) upon the cylindrical insulator surface. This plasma layer is accelerated by electromagnetic forces along the electrodes. After reaching the electrode ends it undergoes radial collapse towards the z -axis and it forms a DMP column (called the PF). In the typical PF discharge one can easily discern different phases of its development, i.e. the initial breakdown and the current-sheath formation phase, the axial acceleration phase and the final radial-compression phase, as shown in figure 3.

PF-type experiments are often performed with the deuterium gas filling, since many laboratories are oriented on the production of high-temperature deuterium (and tritium) plasma in which nuclear fusion reactions can occur. The main aim of this invited lecture is to describe the different phases of such discharges, leading to the formation of a DMP column and accompanying physical processes. Particular attention will be paid to the main physical issues of research on PF-type discharges.

2. Breakdown and current-sheath formation

The development of a sliding discharge along the insulator surface in the PF-type system and its conversion into the breakdown phase has been investigated in different laboratories for many years. It was proved experimentally that the

breakdown depends on the configuration of the electrodes at the main insulator and on the initial gas conditions [6, 7]. Some examples of the investigated electrode configurations are shown in figure 4.

As regards the influence of initial gas conditions, it was observed that when the initial pressure is too low the discharge can develop in the whole inter-electrode volume, and when the pressure is too high the discharge develops in quasi-radial current filaments [8, 9]. In general, the experimental optimization of the electrode–insulator configuration and gas conditions within PF-type facilities requires numerous time-consuming tests. Therefore, attention was paid to the development of physical models and numerical simulations of the breakdown process. Using continuity equations (valid for plasma as a conducting fluid) and the Poisson relations for the chosen electrode–insulator system, it was possible to perform 2D Monte-Carlo simulations [10]. Some results are shown in figure 5.

A comparison of computed density distributions during the formation of a current sheath with the experimental observations has shown a relatively good agreement. To improve the understanding of the breakdown phase other efforts have also been undertaken [11]. However, taking into consideration the complexity of the current-sheath formation phase an accurate quantitative model is still missing.

One of the important issues is the role of the main cylindrical insulator, and in particular the influence of its material composition and surface structure on the development of PF discharges. It has been proved experimentally that the replacement of the traditional insulator made of glass by one made of technical ceramics can considerably improve the operation of the PF facility. Such a change facilitates the breakdown even at the presence of some heavy gas admixtures and it ensures better fusion-neutron yield.

As regards the insulator surface, it was observed in many experiments that some training discharges are needed to achieve so-called ‘good shots’. Probably, the fine structure of the insulator surface, e.g. the absorption and re-absorption of working gases as well as the formation of metal crystallites upon the insulator, play an important role in the breakdown process. Unfortunately, such processes can hardly be taken into account in theoretical modeling. One can conclude that to verify this hypothesis new active experiments with preliminary modifications of the insulator are needed. In fact, such experiments were proposed by the author several years ago [12], but they have not been performed so far. It should be noted that the application of gas puffing near the insulator surfaces has also been considered [13].

In summary of this point one can conclude that further experimental and theoretical studies of the breakdown phase are still needed.

3. Axial acceleration phase

The next phase of a PF-type discharge, i.e. the axial acceleration of the current sheath, can be analyzed on the basis of numerous experimental and theoretical studies. The most popular and effective model is the 2-fluid MHD approach,

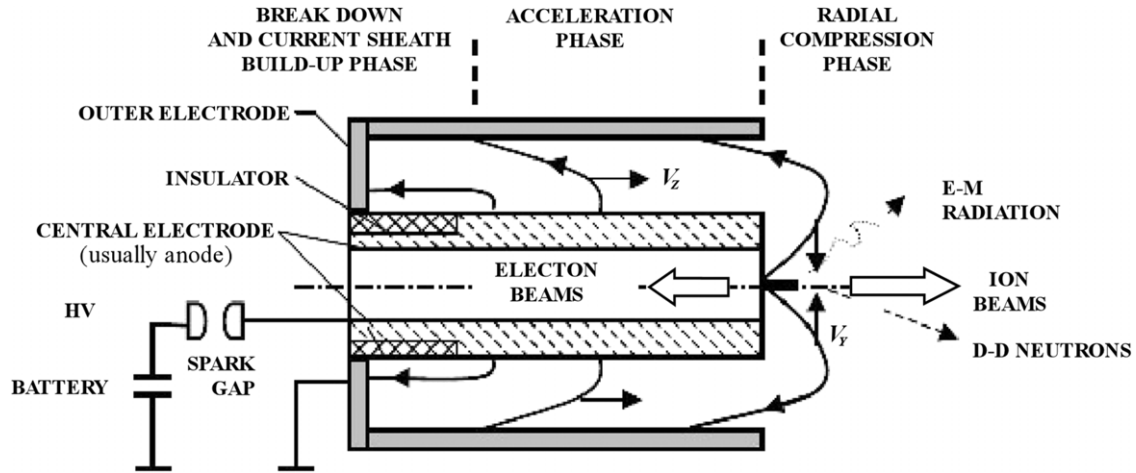


Figure 3. Schematic of the PF-type experiment, which shows different phases of the discharge leading to the formation of the DMP column at the electrode outlet.

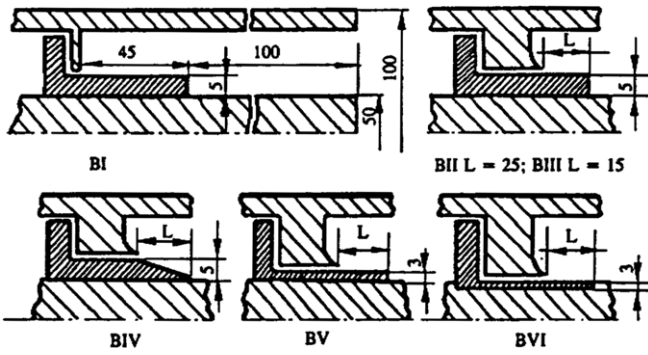


Figure 4. Electrode-insulator configurations studied in a small PF-type device. In configurations BIV to BVI the length L was 15 mm. In contrast to BV, in the configuration BVI the insulator was embodied in the inner electrode. The best configuration was BVI [7].

which uses equations of fluid mechanics and electro-dynamics simultaneously. In particular, use is made of the equation of plasma continuity, equations of momentum and energy balance as well as the Maxwell equations and those characterizing the considered electrical circuit.

The computer simulation must of course be specified for the chosen electrode configuration and gas conditions. Such a modeling is also sensitive to the kinetics of the ionization and transport phenomena. In that case use is often made of so-called Braginski transport coefficients. The ionization rate and recombination might be described by a formula

$$\frac{dn_e}{dt} = n_e(n_0 - n_e) \cdot S - \alpha_r n_e^2 - \beta_{3B} n_e^3,$$

where dn_e/dt is the temporal change of the electron concentration, n_0 is the neutral gas concentration, S is a factor describing electron-ion collisions, α_r is the coefficient of the radiative recombination and β_{3B} is the coefficient of three-body Auger recombination [14]. Since the processes considered depend strongly on the electron temperature and concentration it is impossible to solve this equation with the algorithm used for the MHD equations, because the numerical stability would

require very small time steps and some simplifying procedures must be applied.

An anomalous resistivity of plasma, which appears during PF discharges, should also be taken into account. The assumed boundary conditions must describe the geometry of the PF experiment correctly. The numerical modeling can then deliver information about distributions of the electron temperature and concentration in different instants. An example of computations performed on the basis of the described MHD model is shown in figure 6.

The detailed analysis shows that the ionized gas between coaxial electrodes is pressed towards the inner surface of the external electrode (usually cathode) and it is simultaneously accelerated along the z -axis. The joule heating and thermal conduction losses in such a compressed plasma sheath can be estimated [16].

The total dissipated energy, which is proportional to the contact area of the plasma and the coaxial electrode, can also be estimated, but in the 2D approach no simulation can be performed for the multi-rod (squirrel-cage) cathode configuration. At first glance, one can assume that when the contact area is decreased considerably the energy dissipation by the electron ablation might become dominated. In order to reduce this dissipation and to achieve higher acceleration efficiency it is necessary to understand the interaction of a plasma sheath and electrodes, but it requires more advanced theoretical and experimental studies. Unfortunately, there is no evident progress in experimental studies of the axial acceleration phase. No new experimental data on this phase are available, although some probe and spectroscopic measurements are sometimes undertaken. The main issues of the considered phase seem to be the effects of non-uniformities and quasi-radial filaments, which might be formed and preserved during the axial motion of the current sheath.

4. Radial collapse phase

The third phase of PF-type discharges, which occurs when the current sheath layer reaches the electrode ends and undergoes the radial compression (collapse) leading to the formation of

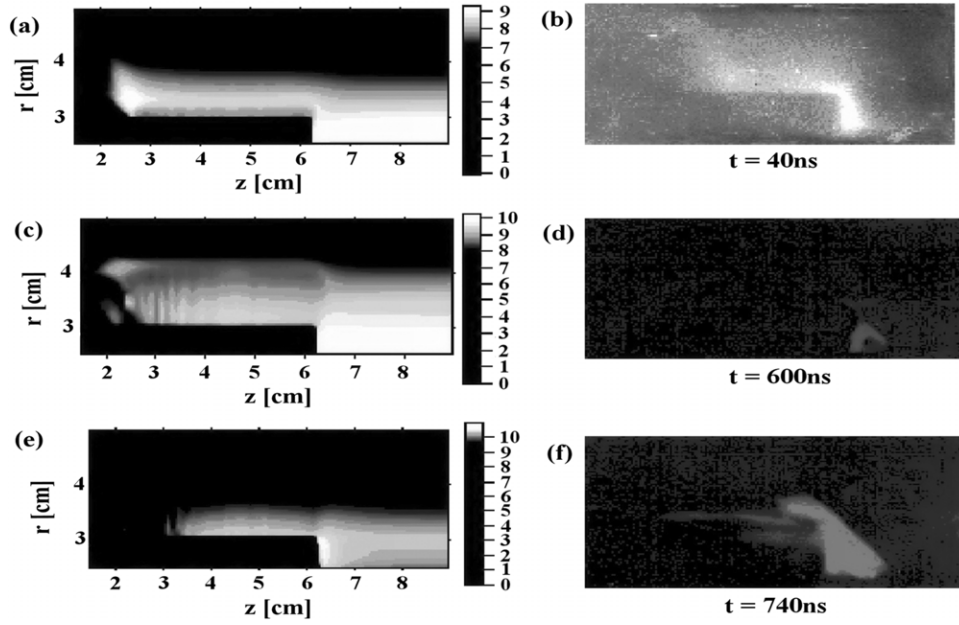


Figure 5. Computed plasma density distribution at different instants of the breakdown and the visible-radiation pictures taken with a high-speed camera side-on the electrode system [10]. In all the diagrams and pictures the upper edge corresponded to the outer electrode (cathode), the bottom edge corresponded to the inner electrode, while the insulator is visible as the bottom-left rectangle.

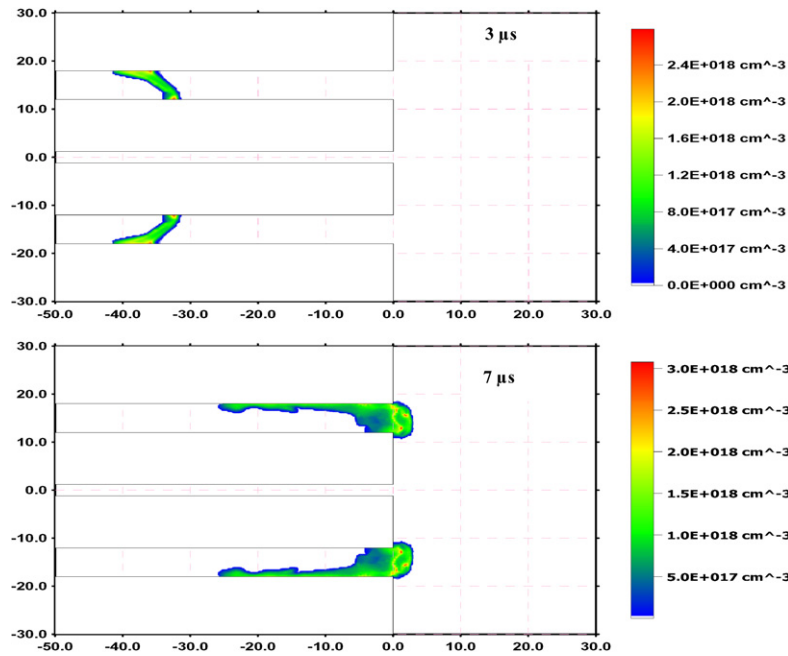


Figure 6. Plasma density distribution in the accelerated current-sheath, as computed for instants 3 and 7 μ s after the discharge beginning in the PF-1000 facility [15]. The tubular ceramic insulator shielded only the first 10 cm part of the inner electrode (not shown in the diagram).

a DMP column, has been extensively investigated in many laboratories. An example of experimental observations, which were performed during this phase, is shown in figure 7.

One can easily see that during the radial collapse of the current sheath a development of some MHD instabilities might be observed, but the maximum compression and the formation of DMP pinch column is achieved if the experimental conditions are favorable. After the maximum compression the distinct instabilities can destroy the DMP column very quickly.

For the theoretical analysis one can also apply the MHD approach, but such a modeling of the collapse phase is

efficient only until the maximum compression. Later on the development of different plasma instabilities requires more sophisticated approaches. Nevertheless, using the extended MHD model described above, some valuable computer simulations of the collapse phase were carried out, as shown in figure 8.

Important information about DMP pinch characteristics can also be gained from measurements of x-rays, as well as fast electron beams (emitted mainly in the upstream direction) and energetic ion beams (emitted mostly along the discharge axis). These electromagnetic and corpuscular emissions can

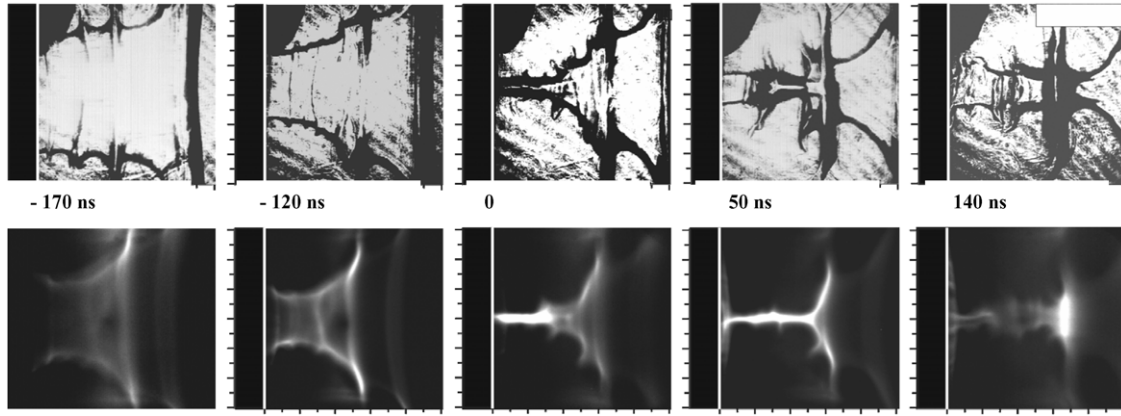


Figure 7. Schlieren images and visible-radiation pictures taken with a high-speed camera before and after the maximum compression ($t = 0$) within the PF-1000 facility [17].

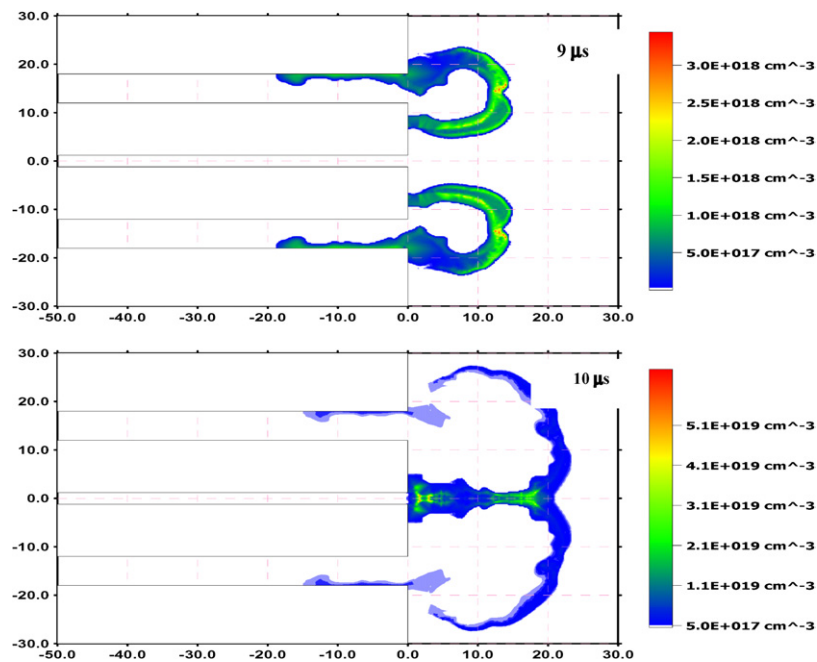


Figure 8. Plasma density distribution during the radial collapse phase, as computed for instants 9 and 10 μs after the discharge beginning in the PF-1000 facility [17]. The tubular ceramic insulator shielded only the first 10 cm part of the inner electrode (not shown in the diagram).

be investigated by means of different diagnostic methods, including time-integrated techniques and time-resolved ones. Particular attention is often paid to the emission of fast neutrons and protons originating from nuclear fusion reactions, which occur within high-temperature DMP produced by discharges in a deuterium gas [18, 19]. An example of the correlation of time-resolved signals corresponding to different emissions is shown in figure 9.

The main issues of such studies are the characteristics of the observed emissions as well as their correlations. Therefore, detailed measurements of time-integrated and time-resolved signals are carried out in many DMP experiments. Particular attention is paid to the appearance of plasma filaments and so-called ‘hot spots’, the emission of very fast electron beams as well as very energetic ion beams and fusion products (fast protons and neutrons from D–D reactions).

5. Appearance of plasma filaments and hot spots

The high-speed camera pictures of the visible radiation, which were taken along the electrode axis in different PF discharges, showed that the current sheath layers can contain distinct quasi-radial filaments, as shown in figure 10.

Such plasma filaments can also exist when the plasma layer is blown out the inter-electrode gap and is subject to the radial compression. It is estimated that the peak current in a single filament can reach several hundred kiloamps, and the local plasma density can amount to 10^{17} – 10^{19} cm^{-3} . Different models of the formation and interaction of the current filaments were proposed, and some authors claimed that the primary filaments constitute filamentary vortices colliding and annihilating during the radial compression [20].

Such processes should lead to the formation of a homogeneous plasma column, but x-ray pinhole pictures taken

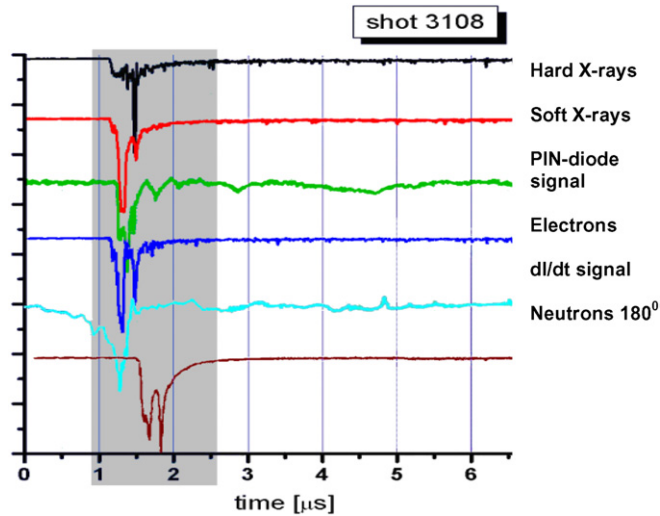


Figure 9. Signals corresponding to hard x-rays, soft x-rays, visible radiation (recorded with a PIN diode), fast electrons, dI/dt waveform and fusion-produced neutrons (recorded at 180°) within the PF-1000 facility operated at 33 kV/734 kJ, with the peak current of 1.7 MA.

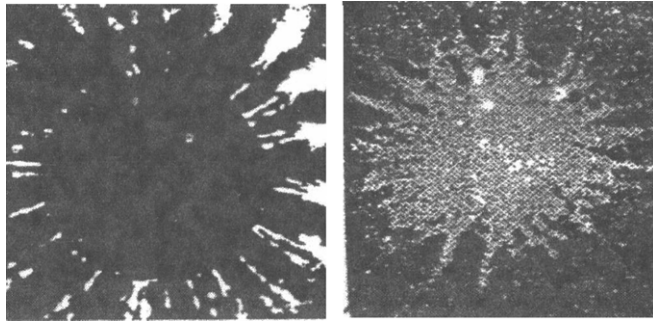


Figure 10. High-speed camera pictures (end-on a PF discharge), which show quasi-radial filaments in a plasma layer during its axial motion and its radial collapse.

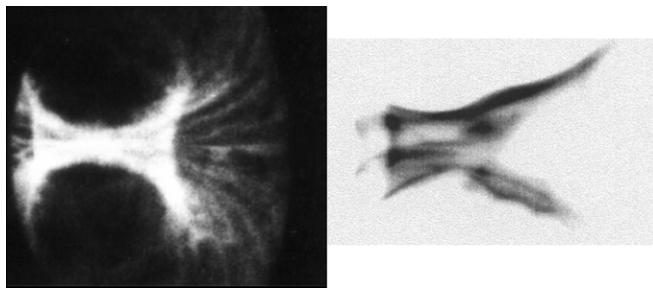


Figure 11. High-speed camera pictures, as taken in two different PF-type experiments, which show distinct plasma filaments during the radial collapse phase and quasi-axial filaments inside the DMP column.

side-on showed that some filaments can exist even during the final compression of the DMP column [21]. The appearance of quasi-axial filaments inside the DMP column was also observed by means of high-speed visible-radiation cameras [22, 23], as shown in figure 11.

It should be noted that similar effects of the plasma filamentation were also observed in Z-pinch experiments, as shown in figure 12. The important issue is the influence

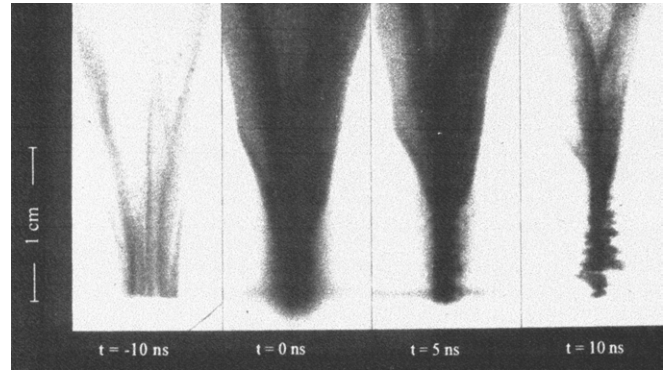


Figure 12. Quasi-axial filaments in the DMP column of a vertical Z-pinch discharge [24].

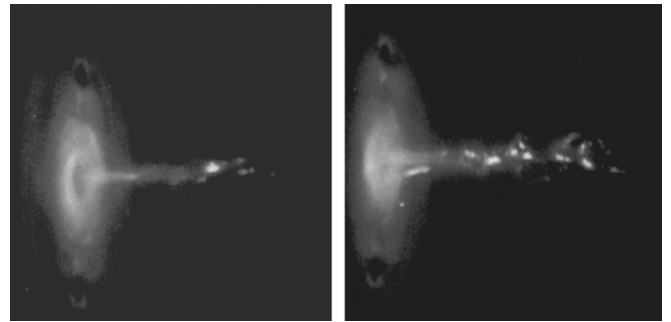


Figure 13. Appearance of miniature hot spots inside the DMP column, which were observed in a medium-scale MAJA-PF experiment [25].

of such filaments on DMP pinch dynamics and its emission characteristics.

In numerous PF (as well as Z-pinch) experiments, particularly in those performed with a heavy gas admixture, inside the DMP column there was also observed the formation of miniature regions of an increased x-ray emission (called hot spots). Some examples are shown in figure 13.

One might suspect that such hot spots are formed by strong local MHD instabilities, which can split the current filaments into small pieces, but the distinct hot spots were observed in many experiments, in which the primary filaments were not recorded at all. Therefore, the mechanism of the hot spot formation remains an open issue.

In order to determine plasma parameters inside the hot spots use was made of various x-ray spectrometers. A detailed analysis of x-ray spectral lines emitted from the hot spots, particularly in shots performed with heavy gas (argon) admixtures, was carried out by means of two identical crystal spectrometers situated side-on but with mutually perpendicular dispersion planes. Those measurements showed that the observed x-ray lines often have different polarization, as shown in figure 14.

The described effect can be explained by the influence of fast (supra-thermal) electron beams, which might be accelerated in strong local electrical fields appearing in DMP [27].

On the basis of the recorded x-ray lines, it was estimated that the local electron concentration can amount to 10^{20} – 10^{21} cm^{-3} and the local electron temperature can exceed 1 keV.

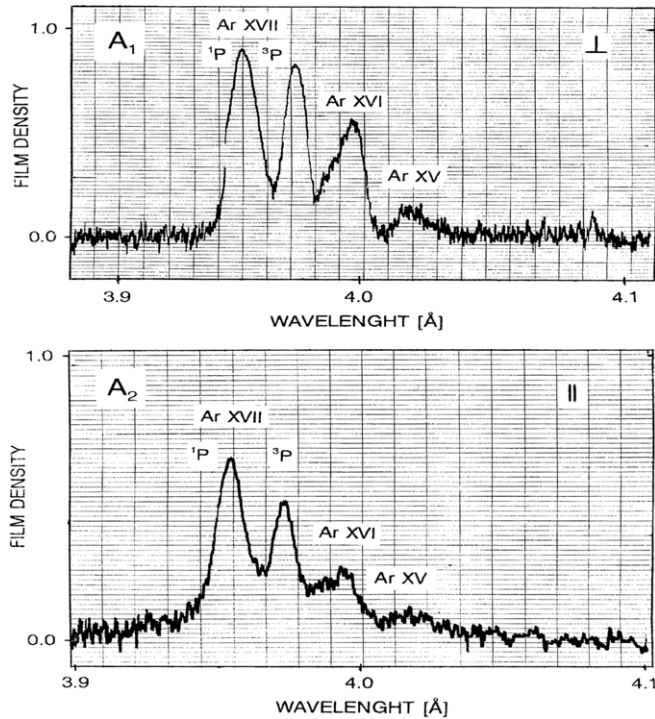


Figure 14. X-ray spectra of highly-ionized argon ions, as recorded in the MAJA-PF experiment [26], which show that polarization of the observed lines is different.

These values were rough estimates, because a small population of supra-thermal electrons could change those parameters by order of magnitude. On the other hand, fast electron beams were recorded at different angles to the z -axis, and differences in the polarization of various x-ray lines can be explained by electrons moving in various directions. It should be noted that, in spite of some theoretical efforts, a quantitative analysis of the polarization effects still remains an unsolved issue.

6. Emission of fast electron beams and accelerated primary ions

A simple phenomenological model of DMP phenomena assumes that the internal current filaments can undergo MHD instabilities, and the local breach points then become the hot spots. Estimations of local electrical fields, which appear due to the separation of electrical charges when a dense plasma filament is broken inside the DMP column, give values of the order of 50 MV cm^{-1} . Such high electric fields can explain the emission of fast electron beams and ions (in opposite directions) from the hot spots constituting the micro-sources.

In fact, the pulsed electron beams emitted from DMP discharges have been observed in many experiments [28, 29]. Time-integrated measurements of such e-beams were performed with magnetic analyzers placed behind the axial channel in the central electrode (anode) of the PF facility [30]. It was shown that (at the charging voltage equal to about 32 kV) the measured electron energy spectrum extends from about 60 keV to about 600 keV. Low-energy electrons can of course be considered as secondary ones, since the DMP column and the working gas (filling the chamber) can be penetrated by

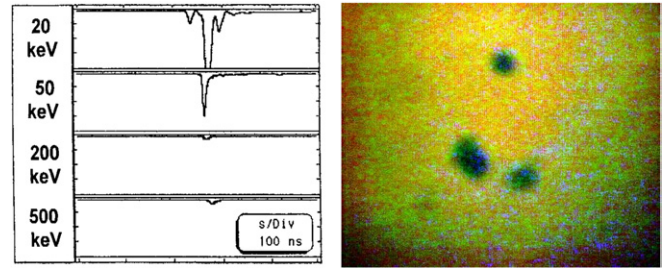


Figure 15. Time-resolved pulses of electrons recorded behind the collector of the MAJA-PF facility and time-integrated images of fast ion beams recorded along the discharge axis, which corresponded to the x-ray pinhole image of the hot spots shown in figure 13 (left).

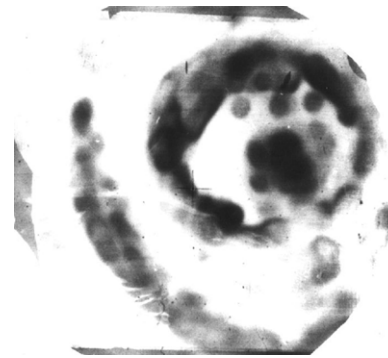


Figure 16. Pinhole image of fast deuteron beams (with $E_d > 380 \text{ keV}$) emitted from a small 20 kJ DMP experiment. The concentric rings of the ion tracks were caused by deflections of the ions in the azimuthal magnetic field surrounding the DMP column.

energetic electrons only. Other measurements, as carried out with witness plates, showed that the fast electrons are mostly emitted in the upstream direction, forming relativistic electron beams (REBs).

The emission of REB was also confirmed by time-resolved measurements performed by means of miniature scintillation detectors placed behind the magnetic analyzer as well as by direct measurements with Cerenkov-type detectors. It was shown that electrons of lower energy ($< 150 \text{ keV}$) are emitted during the period lasting 50–300 ns in the radial collapse phase, while the high-energy electron beams ($> 150 \text{ keV}$) are emitted in short pulses (lasting 10–30 ns) during the first compression of DMP and after its disruption. Simultaneously with the REBs one can also observe the emission of fast ion beams directed in the opposite direction along the discharge axis. These corpuscular emissions are well correlated with the appearance of distinct hot spots, as shown in figure 15.

The pulsed ion streams, which are emitted mostly along the PF-system symmetry axis, have been investigated for many years [31, 32]. Measurements of the ion emission, which were performed within different DMP experiments, demonstrated that the accelerated primary ions (usually deuterons) are emitted in bunches of narrow micro-beams of different energies, forming a complex spatial structure, as shown in figure 16.

The experimental studies of the ion emission, which were carried out with different diagnostic methods, e.g. activation techniques and Thomson spectrometers, showed that a small

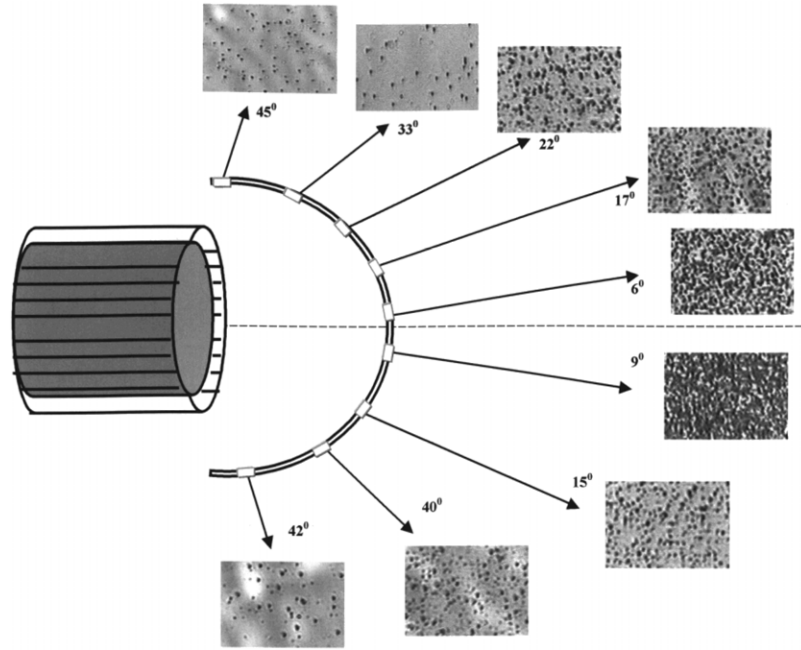


Figure 17. Results of the time-integrated measurements of the ion angular distribution within the PF-1000 facility [35].

1 kJ DMP discharge can deliver about 10^{14} deuterons of energy above 330 keV, while a larger 75 kJ experiment can emit about 10^{15} such deuterons. It was also shown that the PF facility, supplied from a 250 kJ condenser bank charged up to about 50 kV, can generate deuteron beams of energy spectrum ranging up to about 6 MeV. Analogous measurements were also performed for heavy impurity and admixture ions. The high energies of the ion beams, which are evidently emitted from the small sub-millimeter sources (close to hot spots), show that the local electrical fields in DMP might be of the order of 50 MV cm^{-1} . Time-resolved ion measurements showed that the described micro-beams have the FWHM equal to 2–8 ns [33] and their total power can reach 10^{11} – 10^{12} W.

In order to describe the emission of fast ions different theoretical models were proposed, and in particular the acceleration by very strong local electric fields (see above) generated by DMP instabilities, wave-particle interactions, multiple reflections of ions from the collapsing current sheath, etc. One of the most popular models assumes the appearance of so-called plasma diodes, which might be identified with plasma regions situated close to the observed hot spots. In such a model some electrons, accelerated within the plasma diode, can interact with DMP and induce an increased x-ray emission. The primary ions, which are accelerated in the opposite direction, can move through DMP in the forward direction and undergo deflections by local magnetic fields (connected with a flow of discharge currents). Unfortunately, none of the models mentioned above can be used to explain all the characteristics of the ion emission. Relatively good results are obtained if trajectories of medium-energy (10–100 keV) deuterons are analyzed with the so-called gyrating particle model (GMP) proposed several years ago [34]. This model cannot, however, explain all characteristic features of ion angular distributions, and in particular the local minimum of the ion flux observed at the z -axis.

Measurements of the ion angular distributions were performed in different PF facilities, e.g. by means of solid-state nuclear track detectors (SSNTD), as shown in figure 17.

To explain the observed angular distributions of the fast deuterons emitted from PF discharges, it was necessary to take into account the filamentary structure of the DMP column [36]. Numerical modeling of ion trajectories, which was performed for the assumed ‘flower-like’ configuration of several current filaments, gave the ion angular distribution with a local minimum in qualitative agreement with the experimental data, as shown in figure 18.

More detailed computations concerned various numbers and spatial distributions of the current filaments, as well as different local sources (starting points) and initial velocities (energies) of deuterons. It was found that the influence of electrical fields, which might be induced by radial motions of the current filaments, should also be taken into account [37].

Although the filamentary model gives good qualitative agreement with the results of experimental observations, mechanisms of the ion acceleration as well as real motions of the accelerated ions inside DMP still remain unsolved problems. In fact these issues are of importance also for estimates of the yields of nuclear fusion reactions, which occur inside high-temperature deuterium DMP.

7. Emission of fusion-produced neutrons and protons

Fusion-produced neutrons from PF-type deuterium discharges have been studied in plasma laboratories for many years [3,4,30]. The experimental scaling laws describing the neutron yield (Y_n) as a function of the initial energy (W_0) and a pinch current (I_p) were found to be $Y_n \sim W_0^\alpha$ and $Y_n \sim I_p^\beta$, where $\alpha = 2.0$ – 2.2 and $\beta = 3.3$ – 4.4 depend on energy supplied

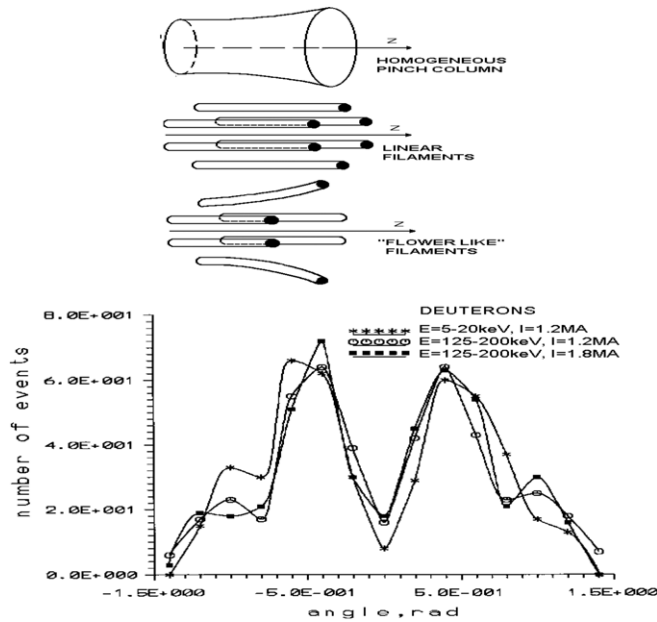


Figure 18. Considered structures of the pinch column used for computing ions trajectories (upper figure) and the angular distributions of deuterons (lower diagram), as computed for different values of deuteron energy and discharge current.

and the facility type. Unfortunately, in all the investigated DMP experiments the saturation of the neutron emission was observed, when the discharge current was increased above a critical value (depending on experimental conditions). In order to increase the neutron yield different efforts were undertaken, and in some cases they enabled the saturation level to be shifted up [4, 38].

Some efforts connected with the optimization of the main insulator and electrodes were described above. Interesting experiments were performed with the application of additional gas (deuterium) targets, which were produced at the electrode ends by means of a fast-acting gas valve, but no distinct improvement of the neutron emission has been achieved. Other tests were directed to the use of fast deuterons, which can escape from the DMP column. For that purpose there were applied special cryogenic targets placed in front of the electrode outlet and coated with a frozen D_2O (heavy ice) layer [39]. Such tests showed that some increase in an average neutron yield could be achieved, but this approach did not eliminate the saturation effect. Numerous efforts to improve the fusion-neutron yield were undertaken within the large PF-1000 experiment [40]. Different sets of the tubular and squirrel-cage electrodes were tested and the operational energy value was increased to about 1 MJ. Unfortunately, the observed neutron yield from the PF-1000 has been saturated at energy above 650 kJ [4]. Recently, new efforts have been undertaken to improve the PF-1000 facility operation and to increase the fusion-neutron yield by modifications of the collector and electrodes as well as by the application of the working gas injection near the main insulator surface. Appropriate tests will be performed during the coming autumn.

The detailed studies of the neutron emission from DMP also included measurements of the neutron emission anisotropy, which were carried out by means of different sets

of silver-activation counters and scintillation detectors. Some examples of time-resolved neutron signals from the PF-360 experiment [41] are shown in figure 19.

Using the time-of-flight (TOF) technique, it was estimated that the energy of neutrons in the first pulse measured along the z -axis was $E_n = (2.65 \pm 0.13) \text{ MeV}$, while that measured side-on was $E_n = (2.47 \pm 0.15) \text{ MeV}$. One should also note that the ratio of the neutron signals, as measured at 10° and 77° for different instants, varied considerably. The experimental evidence showed that the anisotropy of the neutron emission in the successive pulses was slightly different. That might be a result of different mechanisms involved in the neutron production, e.g. beam-target interactions, thermonuclear reactions etc. In fact, the role played by these mechanisms still remains an important issue.

Important information about the localization of micro-regions of fusion reactions, which occur within the DMP column, can be gained from investigation of charged products of D-D reactions, and particularly from measurements of fusion-produced protons of energies $>3 \text{ MeV}$. In contrast to fusion neutrons, the fusion-produced protons are deflected by local magnetic fields inside and outside the DMP region. Therefore, they can deliver some information about such fields and the DMP micro-structure. Preliminary measurements of fusion protons were carried out within PF-360 and PF-1000 facilities [42, 43].

To record fusion protons use was made of miniature pinhole cameras, which were placed at different angles to the discharge axis. Those cameras had input diaphragms of 3 mm in diameter and were equipped with PM-355 track detectors shielded by Al foil-filters $80 \mu\text{m}$ in thickness, which eliminated heavier ions and protons of energies $<3 \text{ MeV}$. It enabled the recording of the fusion-produced protons, as shown in figures 20 and 21.

An analysis of the recorded proton images showed that the fusion protons are emitted from miniature sources placed inside the DMP region. It confirmed the hypothesis that the DMP column, and not the surrounding gas, is the main source of the fusion products. The proton emission was strongly anisotropic, which confirmed the important role of local magnetic fields induced by currents flowing through the DMP column. Similar results were obtained within the PF-1000 experiment, but the angular anisotropy of the proton emission was stronger, which might be explained as a result of higher discharge currents and stronger magnetic fields.

A quantitative analysis of proton tracks, which was based on the calibration diagrams of the applied track detectors, showed that histograms of the tracks diameters recorded in the both experiments (PF-360 and PF-1000) were similar, as shown in figure 22.

Recent experimental studies examined the influence of the filamentary currents flowing through the DMP column (see above) and induced azimuthal magnetic fields on the emission of the fusion protons. In order to confirm the filamentation of the DMP column, several pinhole cameras were placed in the PF-360 facility at different azimuthal angles around the z -axis. If the DMP column were a homogeneous one,

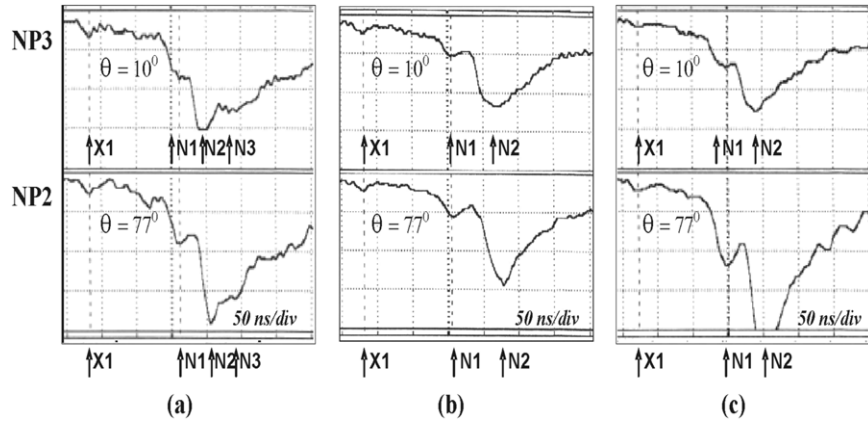


Figure 19. Neutron signals from scintillation probes located at different angles, at a distance of about 3 m from the electrode outlet, as recorded in PF-360 at different operational conditions: (a) $U_o = 29$ kV, $E_o = 106$ kJ, $p_0 = 6.25$ hPa D_2 , and $Y_n = 2.16 \times 10^{10}$; (b) $U_o = 30$ kV, $E_o = 113$ kJ, $p_0 = 7.95$ hPa D_2 , and $Y_n = 1.41 \times 10^{10}$; (c) $U_o = 30$ kV, $E_o = 113$ kJ, $p_0 = 7.8$ hPa D_2 and $Y_n = 1.54 \times 10^{10}$. The reference point is the x-ray signal (X1) and neutron pulses are denoted by N.

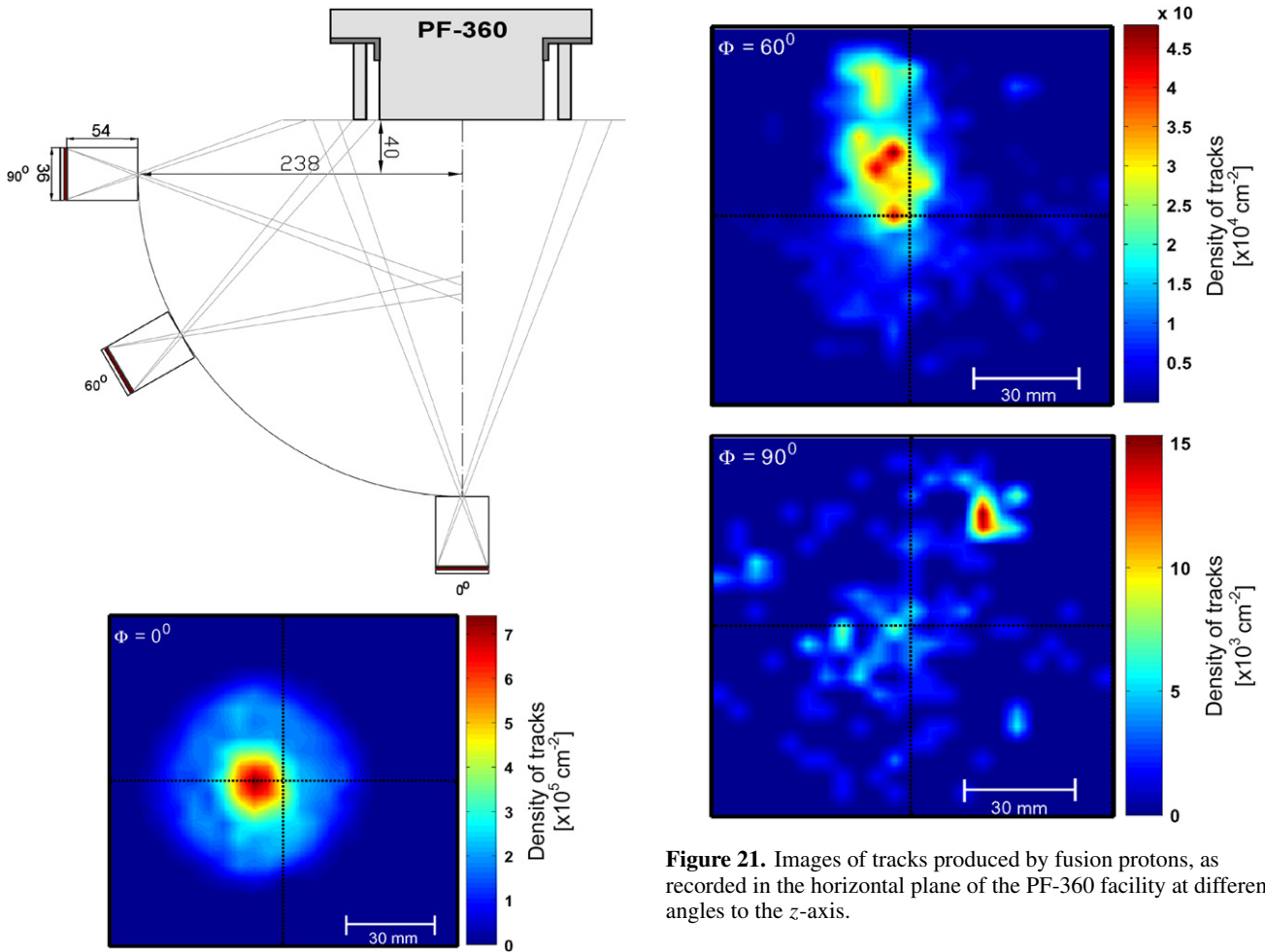


Figure 20. Arrangement of pinhole cameras in the horizontal plane of the PF-360 facility, and an exemplary image of tracks produced by fusion protons emitted along the z -axis.

those cameras should record almost identical pinhole images of fusion protons emitted in different radial directions. The experimental measurements, carried out with four pinhole cameras at the PF-360 facility [44], showed that the recorded pictures differ considerably, as shown in figure 23.

Figure 21. Images of tracks produced by fusion protons, as recorded in the horizontal plane of the PF-360 facility at different angles to the z -axis.

The observed azimuthal anisotropy of the fusion-proton emission can be explained by current filaments and local magnetic fields, which appear within the DMP column and influence proton trajectories. In order to verify this hypothesis more detailed measurements by means of many pinhole cameras, as well as appropriate numerical computations are needed.

A quantitative analysis of the proton images, recorded in the horizontal plane and the plane vertical to the z -axis, enabled

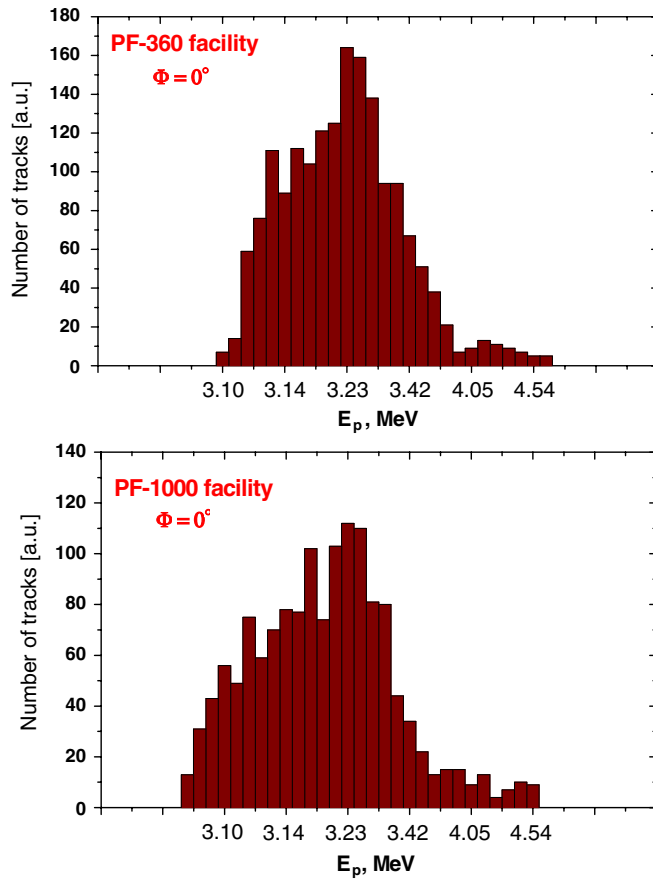


Figure 22. Histograms of the recorded tracks, which show that energy distributions of the fusion protons in the compared experiments were very similar [44].

the angular distributions of fusion protons to be determined, as shown in figure 24.

The presented results constitute the experimental evidence that the micro-structure of the DMP region very strongly influences the emission of fusion protons. On the basis of the measurements performed with the pinhole cameras one can compute proton trajectories outside the DMP column, but an analysis of proton motions inside the DMP region still remains an unsolved problem.

8. Summary and conclusions

The most important results of the studies presented in this lecture can be summarized as follows:

1. The PF facilities are relatively simple and inexpensive devices, which can produce DMP of nuclear fusion interest, emitting intense electromagnetic and corpuscular pulses as well as D–D fusion reaction products, i.e. fast neutrons and protons.
2. The 2D simulations of the breakdown phase enabled the electron and ion concentrations to be computed, but the Joule heating, thermal conduction loss and total energy dissipation require more detailed theoretical analysis. An important issue is the role of the main cylindrical insulator, and in particular its material composition and surface structure on the development of plasma discharges.

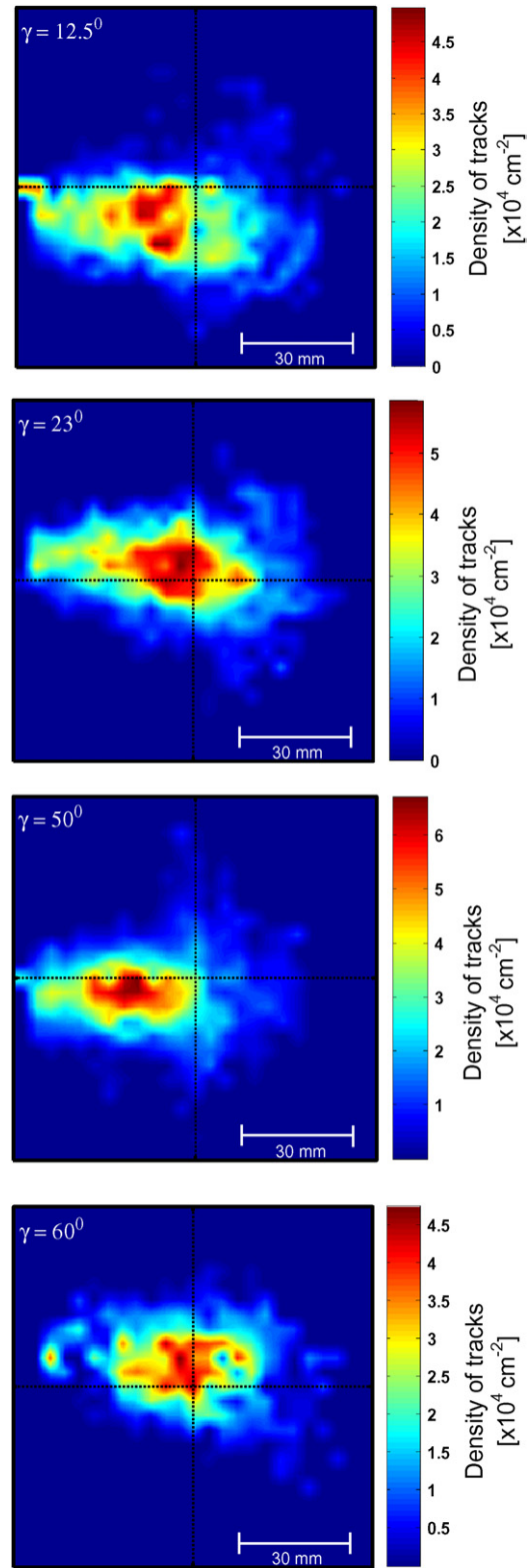


Figure 23. Images of fusion-proton tracks recorded at different azimuthal angles around the DMP column (i.e. in the plane vertical to the z -axis) within the PF-360 facility.

3. The axial acceleration phase of PF-type discharges is relatively well described by 2D MHD models, although new experimental data are still missing. The main issue of this phase seems to be the influence of non-uniformities

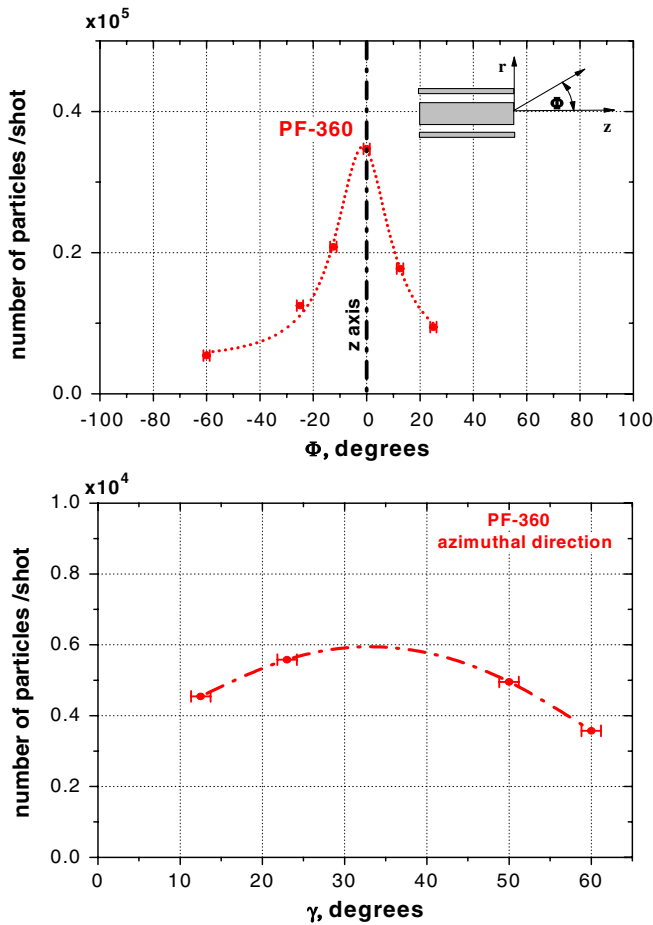


Figure 24. Angular distribution of the fusion protons in the horizontal plane (above) and that in the plane vertical to the z -axis (below), as measured in the PF-360 experiment.

and quasi-radial filaments formed and preserved during the axial motion of the plasma sheath.

4. The radial collapse phase of PF discharges can partially be described by an extended MHD approach, but developments of different plasma instabilities and analyses of their role require sophisticated models and detailed experimental studies of the electromagnetic and corpuscular emissions from the DMP column. The main issues of these studies are characteristics and correlations of the observed radiation pulses.
5. Research on dynamics of the DMP column and its internal structure have already delivered valuable information about current filaments and hot spots, but more accurate space- and time-resolved measurements are needed. Mechanisms of the formation of the filaments and hot spots remain open issues. Another unsolved problem is a quantitative analysis of the polarization of the increased x-ray emission from the hot spots.
6. Experimental studies of fast electron beams and fast ion streams, which are emitted from the DMP column, delivered information about their spatial and energetic characteristics, but more accurate correlation measurements are needed. Also needed is further theoretical analysis of the electron and ion behavior inside and nearby the DMP column. Mechanisms of

the electron- and ion-acceleration as well as real motions of the charged particles inside the DMP region remain unsolved problems. In fact these issues are of importance also for estimates of nuclear fusion yields.

7. The emission of fusion-produced neutrons was investigated in different DMP experiments and numerous efforts were undertaken to optimize the fusion-neutron yields, e.g. by an increase in the discharge current, the use of additional gas- and solid-targets, etc. The emission of fusion-produced protons has recently been investigated in two different PF facilities, and valuable information about the influence of filamentary currents and induced magnetic fields in the DMP columns has been obtained. The issue remains an analysis of proton motions inside the DMP region.

In summary, one may conclude that further theoretical and experimental studies of physical phenomena occurring in the DMP columns are necessary in order to solve the main issues described above.

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